

Data Cleaning & Essential Functions Assignment – Rakesh Ponnoju

Drive Link -

1. What is data cleaning, and why is it important in data analysis?

Data cleaning is the process of identifying and correcting errors, missing values, duplicate records, and inconsistent formatting in a dataset.

Importance of Data Cleaning:

- Improves data accuracy
 - Helps in better decision-making
 - Removes duplicate or incorrect data
 - Makes analysis reliable
 - Saves time during reporting and visualization
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What are the potential consequences of analyzing unclean or messy data?

If unclean data is analyzed:

- Wrong conclusions may be generated
 - Reports become inaccurate
 - Duplicate values affect calculations
 - Missing data may distort analysis
 - Business decisions can become unreliable
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Explain the common steps involved in cleaning and organizing data.

Common Steps:

1. Remove duplicate records
2. Correct spelling mistakes
3. Fill or remove missing values
4. Remove extra spaces using TRIM
5. Standardize formatting
6. Validate data entries
7. Sort and filter the dataset
8. Check formulas and calculations

2. How would you sort the dataset first by "Department" (A–Z) and then by "Salary" (Largest to Smallest)?

Employee	Department	Salary
Pranav	HR	5000
Sonu	IT	4000
Rahul	IT	2500

Steps:

1. Select the complete dataset including headings
2. Go to the **Data** tab in Excel
3. Click on **Sort**
4. In the “Sort by” option:
 - Select **Department**
 - Order → **A to Z**
5. Click **Add Level**
6. In “Then by”:
 - Select **Salary**
 - Order → **Largest to Smallest**
7. Click **OK**

3. Explain the use of text functions such as TRIM, LEFT, RIGHT, MID, and CONCAT in data cleaning.

Uses in Data Cleaning:

- **Standardizing text**
- **Extracting important information**
- **Combining multiple columns**
- **Removing unwanted spaces**

C7 ✕ ✓ <i>fx</i> =TRIM(A1)						
	A	B	C	D	E	F
1	Employee	Department	Salary			
2	Pranav	HR	5000			
3	Sonu	IT	4000			
4	Rahul	IT	2500			
5						
6	Function	Purpose	Example			
7	TRIM	Removes extra spaces	Employee			
8	LEFT	Extracts characters from left side	Emplo			
9	RIGHT	Extracts characters from right side	oyee			
10	MID	Extracts text from middle	ploy			
11	CONCAT	Combines text values	EmployeeDepartment			
12						
13						

4. What is the role of date functions like TODAY in managing datasets?

The TODAY() function returns the current date automatically.

Roles of TODAY():

- Tracks deadlines
- Calculates remaining days
- Monitors attendance
- Creates dynamic reports
- Updates date automatically every day

Example:

=TODAY()

Font

Alignment

✓ *fx*

=TODAY()

B	C	D	E	F	G	H
Department	Salary		26-05-2026			
HR	5000					
IT	4000					
IT	2500					
Purpose	Example					

5. Apply Data Validation to Restrict Quantity Values (1 to 10)

D2		✕	✓	<i>fx</i>	
	A	B	C	D	E
1	Customer Name	Product Name	Category	Quantity	Unit Price (\$)
2	Jane Smith	Shoes	Electronics		81
3	Isabella Moore	Laptop	Electronics		121
4	Daniel Davis	Sofa	Clothing		239
5	Alex Moore	Shoes	Electronics		500
6	Michael Johnson	Table Lamp	Home Decor		423
7	Daniel Johnson	Backpack	Electronics		160
8	Isabella Davis	Headphones	Electronics		348
9	Jane Davis	Headphones	Electronics		152
10	Alex Wilson	T-shirt	Home Decor		369
11					
12					

Quantity Input
Please enter a whole number between 1 and 10.

6. Understand and Apply LEFT, RIGHT, MID, and LEN

Function Purpose

LEFT Extract characters from left side

RIGHT Extract characters from right side

Function Purpose

MID Extract characters from middle

LEN Count total characters

- A. Extract first 5 characters from "ExcelTipsAreGreat"
- B. Extract last 4 characters from "DataAnalysis.xlsx"
- C. Extract the substring "Tips" –
- D. Count total characters in "Hello World!"
- E. Extract middle 6 characters from "12345-67890-ABCDE"

ExcelTipsAreGreat	Excel
DataAnalysis.xlsx	xlsx
ExcelTipsAreGreat	Tips
Hello World!	12
12345-67890-ABCDE	67890-

7. Understand how to combine text using CONCAT, TEXTJOIN, and the & operator

a) Use CONCAT to combine "Hello" and "World" with a space in between

Hello World
Hello World Hello World

=CONCAT("Hello", " ", "World") & =CONCAT(A19, " ", B19)

b) Combine "Apple", "Banana", and "Cherry" into a single string separated by commas using TEXTJOIN

Apple	
Banana	Apple, Banana, Cherry
Cherry	

=TEXTJOIN(" ", TRUE, A20:A22)

c) Use the & operator to create the string "2025: Excel Functions"

Formula:

= "2025" & ": " & "Excel Functions"

Output:

2025: Excel Functions

d) Create a comma-separated list from the range A1:A5 using TEXTJOIN

Formula:

=TEXTJOIN(" ", TRUE, A1:A5)

Explanation:

- Combines all values from cells A1 to A5

- Adds commas between them
- Ignores empty cells because TRUE is used

e) Combine first names in column A with last names in column B to create full names in column C

Font

Alignment

8. Understanding TODAY() and NOW()

a) Difference between TODAY() and NOW()

Function	Purpose
TODAY()	Returns current date only
NOW()	Returns current date and current time

Examples

TODAY()
 =TODAY ()

Example Output:
 26-05-2026

NOW()

=NOW ()

Example Output:

26-05-2026 16:56

When to Use

Function	Usage Example
TODAY()	Attendance, deadlines, age calculation
NOW()	Timestamp logging, real-time tracking

	26-05-2026		
	26-05-2026 16:56		

c) Formula using NOW() to display current date and time in format MM/DD/YYYY HH:MM AM/PM

Formula:

=TEXT(NOW(),"MM/DD/YYYY HH:MM AM/PM")

d) What happens when worksheet is reopened the next day?

If a cell contains:

=TODAY()

The value updates automatically to the new current date whenever the workbook is reopened or recalculated.

Example:

- Today → 26-05-2026
- Tomorrow after reopening → 27-05-2026

e) Keyboard shortcut to insert static current date

Shortcut:

Ctrl + ;

Explanation:

- Inserts today's date as a fixed value
- Does not change automatically later